Lipid Biosynthesis

Lipids Fulfill a Variety of Biological Functions

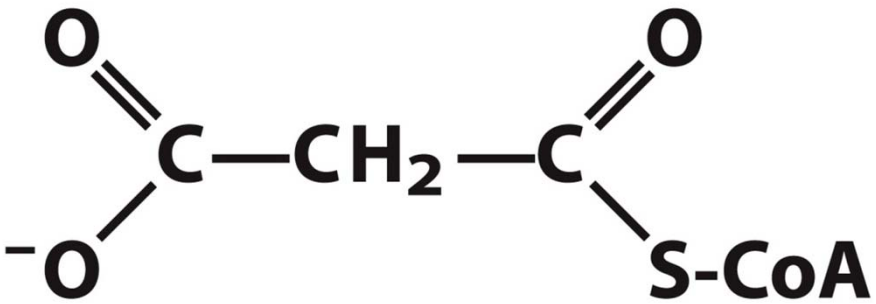
- Energy storage
- Constituents of membranes
- Anchors for membrane proteins
- Cofactors for enzymes
- Signaling molecules
- Pigments
- Detergents
- Transporters
- Antioxidants

Catabolism and Anabolism of Fatty Acids Proceed via Different Pathways

- Catabolism of fatty acids
  - *produces* acetyl-CoA
  - *produces* reducing power (NADH, $FADH_2$)
  - takes place in the *mitochondria*
- Anabolism of fatty acids
  - *requires* acetyl-CoA and malonyl-CoA
  - *requires* reducing power from NADPH
  - takes place in *cytosol* in animals, *chloroplast* in plants

Overview of Fatty Acid Synthesis

- Fatty acids are built in several passes, processing one acetate unit at a time.
- The acetate is coming from activated malonate in the form of malonyl-CoA.
- Each pass involves reduction of a carbonyl carbon to a methylene carbon.



Malonyl-CoA

Unnumbered 21 p811
Lehninger Principles of Biochemistry, Seventh Edition
© 2017 W. H. Freeman and Company

Shuttle for transfer of acetyl groups from mitochondria to the cytosol

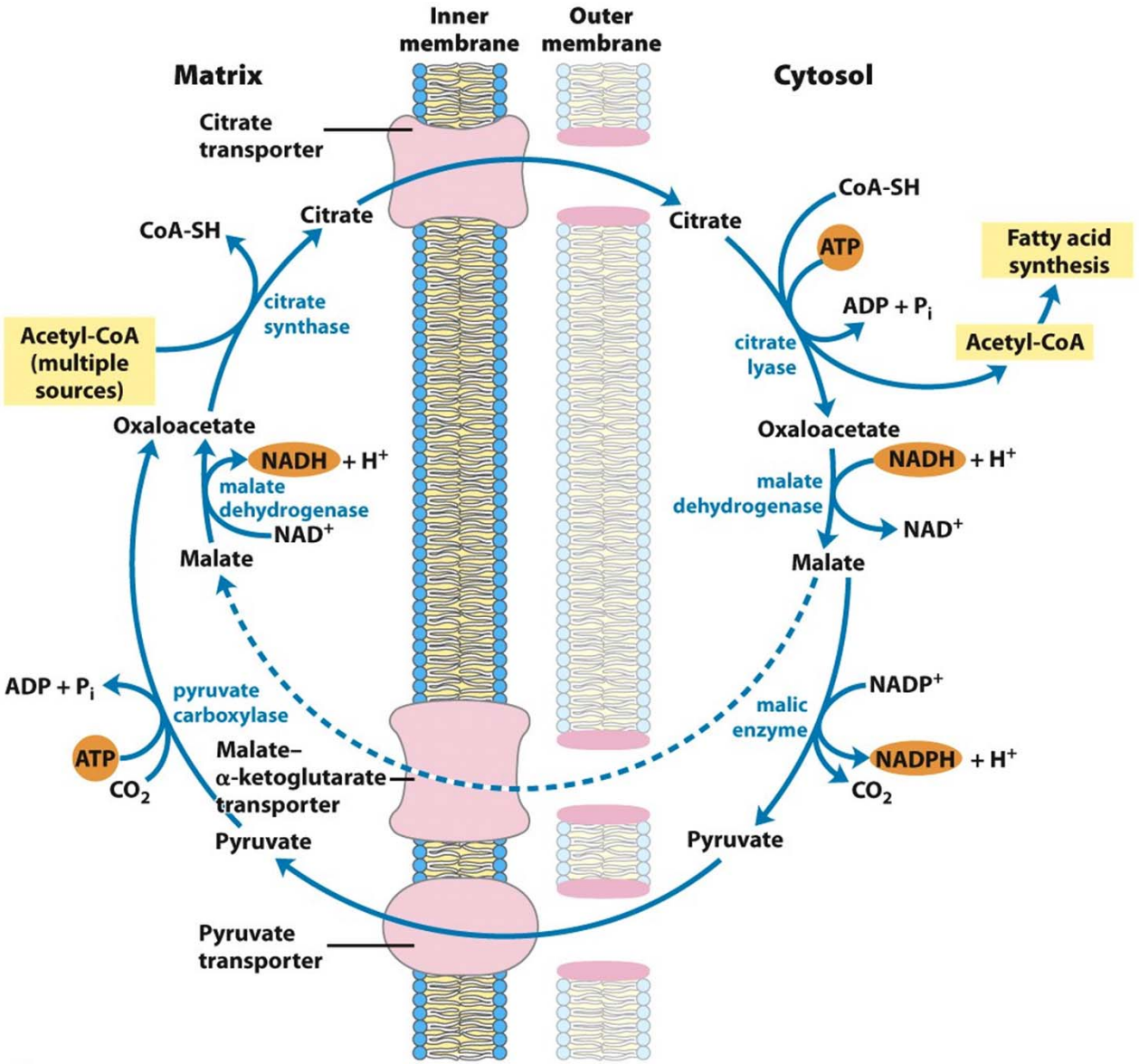


Figure 21-10

Lehninger Principles of Biochemistry, Fifth Edition

© 2008 W. H. Freeman and Company

Fatty Acid Synthesis Occurs in Cell Compartments Where NADPH Levels Are High

- *Cytosol* for animals, yeast
- Sources of NADPH:
  - in adipocytes: pentose phosphate pathway and malic enzyme (Fig. 21-9)
    - NADPH made as malate converts to pyruvate + CO_2.
  - in hepatocytes and mammary gland: pentose phosphate pathway
    - NADPH made as glucose-6-phosphate converts to ribulose 6-phosphate.

Production of NADPH

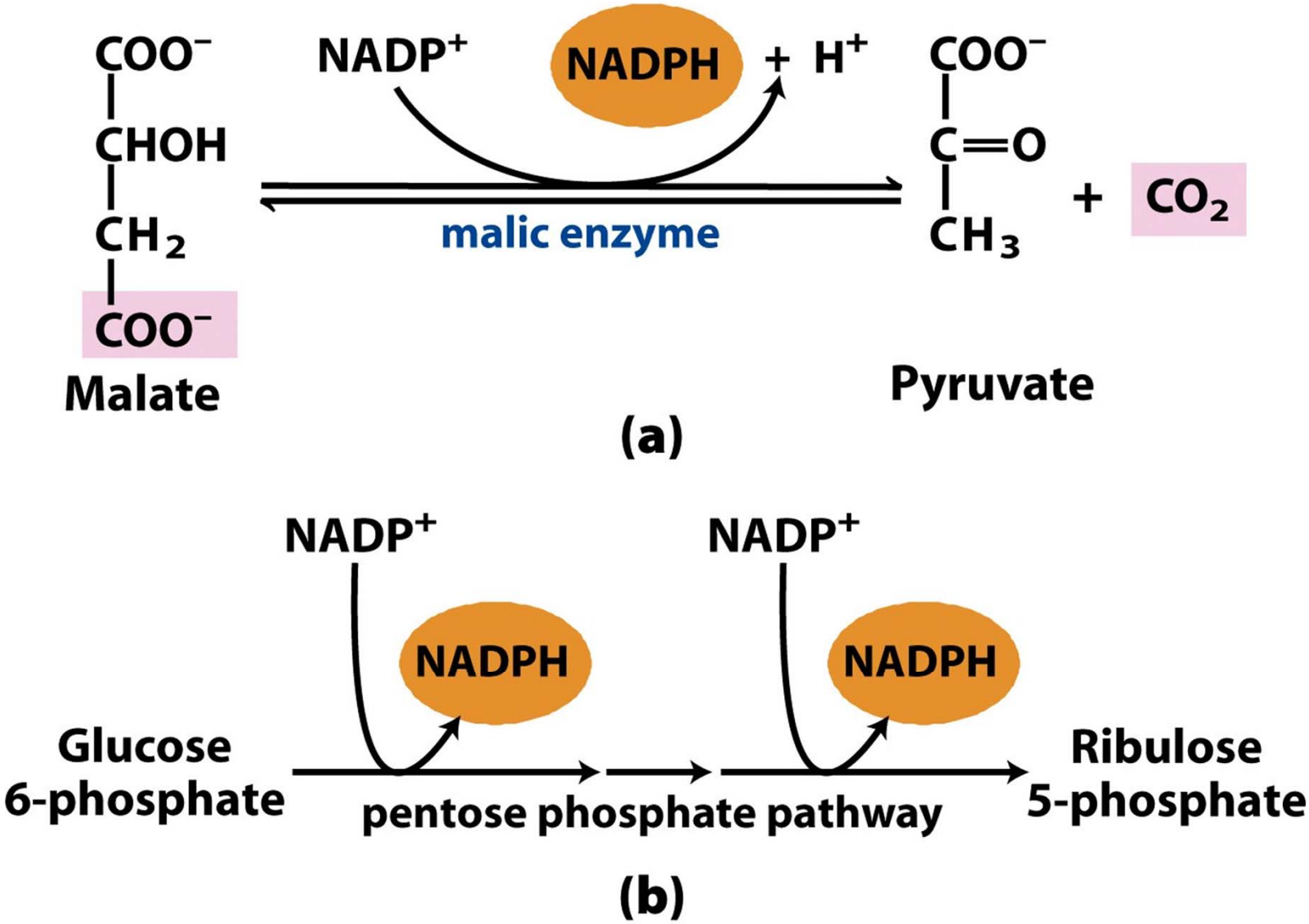
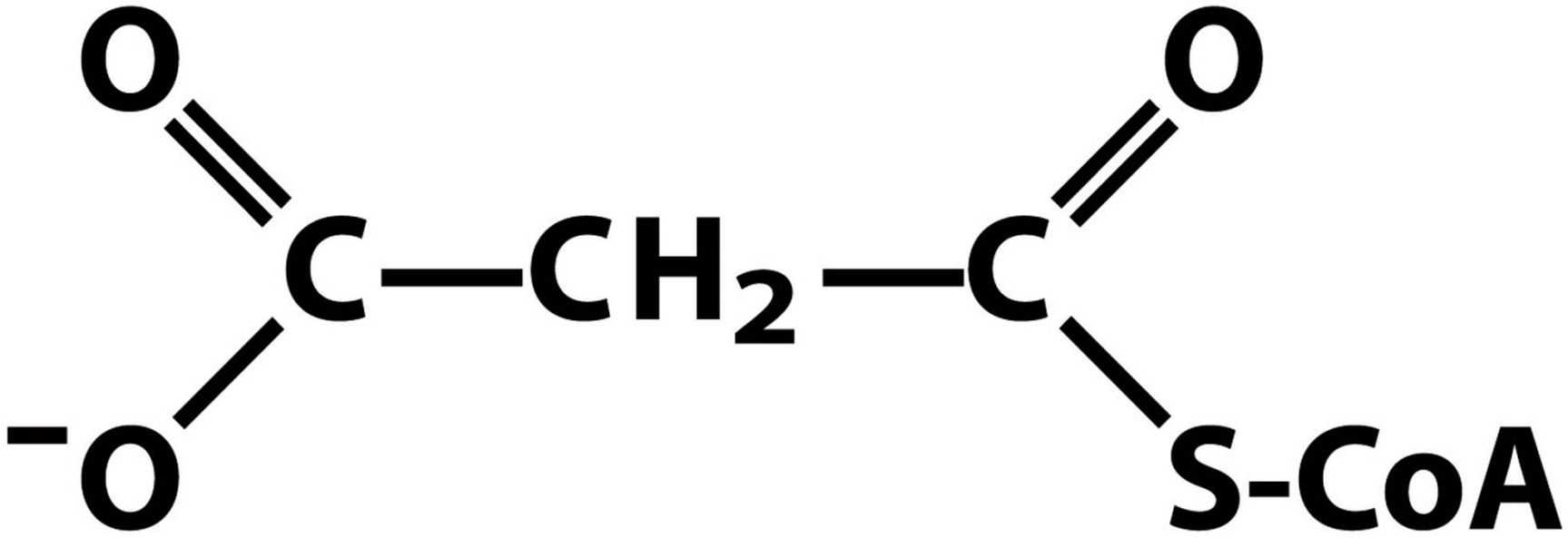


Figure 21-9
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W. H. Freeman and Company

Malonyl-CoA Is Formed from Acetyl-CoA and Bicarbonate

- Reaction carboxylates acetyl-CoA
- Catalyzed by *acetyl-CoA carboxylase*
  - The enzyme has three subunits:
    - One unit has biotin covalently linked to Lys.
    - Biotin carries CO_2.
    - In animals, all three subunits are on one polypeptide chain.
  - HCO_3^- (bicarbonate) is the soluble source of CO_2.



Malonyl-CoA

Unnumbered 21 p805
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W. H. Freeman and Company

The acetyl-CoA carboxylase reaction

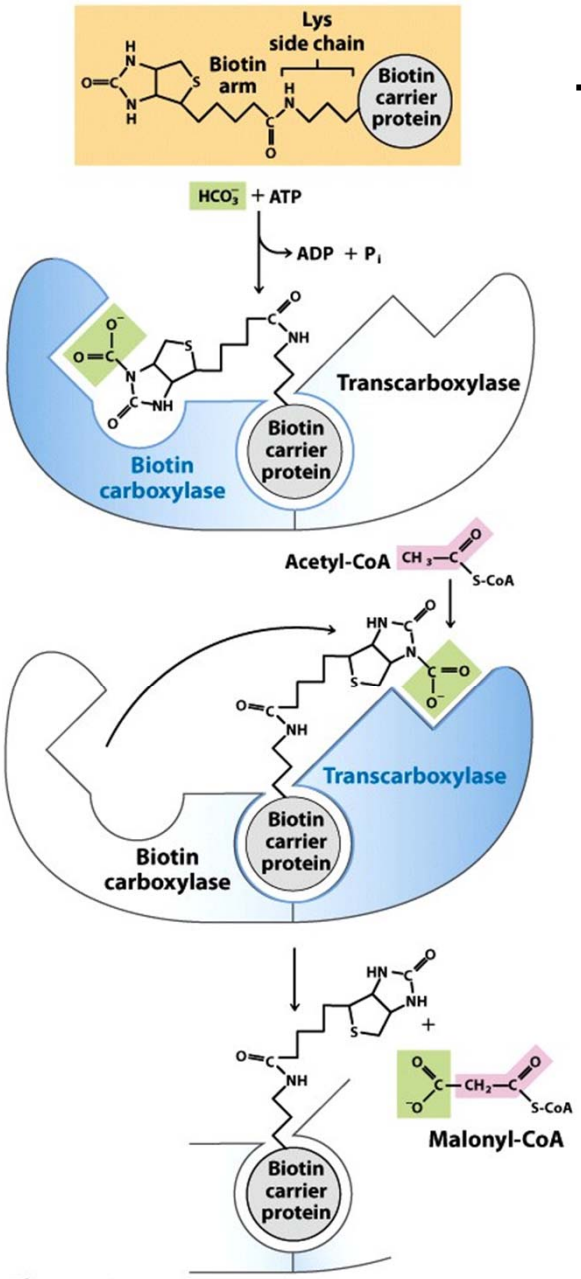


Figure 21-1
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W. H. Freeman and Company

Fatty Acid Synthesis Is Tightly Regulated via ACC

- Acetyl CoA carboxylase (ACC) catalyzes the rate-limiting step.
  - ACC is feedback-inhibited by palmitoyl-CoA.
  - ACC is *activated* by **citrate**.
    - Citrate is made from acetyl-CoA in mitochondria (acetyl-CoA^{mt}).
    - Citrate signals excess energy to be converted to fat.
  - When [acetyl-CoA]^{mt} ↑ is converted to citrate… citrate is exported to cytosol.

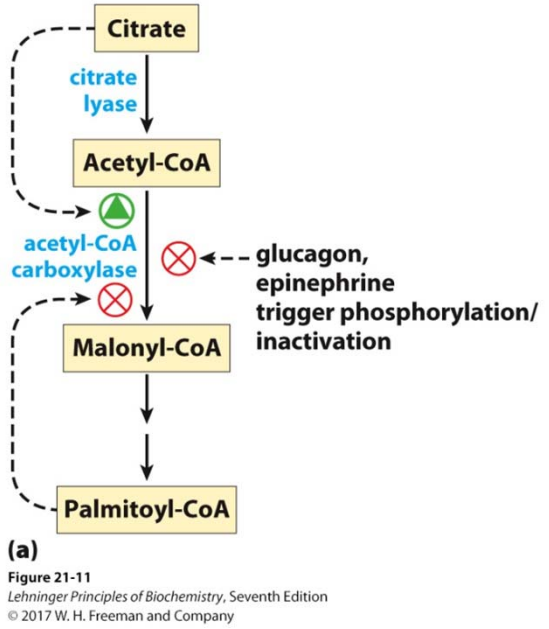


Figure 21-11
Lehninger Principles of Biochemistry, Seventh Edition
© 2017 W. H. Freeman and Company



Courtesy James M. Ntambi, PhD, Professor of Biochemistry, Steenbock Professor of Nutritional Sciences, University of Wisconsin–Madison

ACC Is Also Regulated by Covalent Modification

- Inhibited when energy is needed
- Glucagon and epinephrine:
  - reduce sensitivity of citrate activation
  - lead to phosphorylation and inactivation of ACC
    - ACC is active as dephosphorylated monomers.
    - When dephosphorylated, ACC polymerizes into long inactive filaments.
    - Phosphorylation reverses the polymerization.

Coordinated regulation of fatty acid synthesis and breakdown

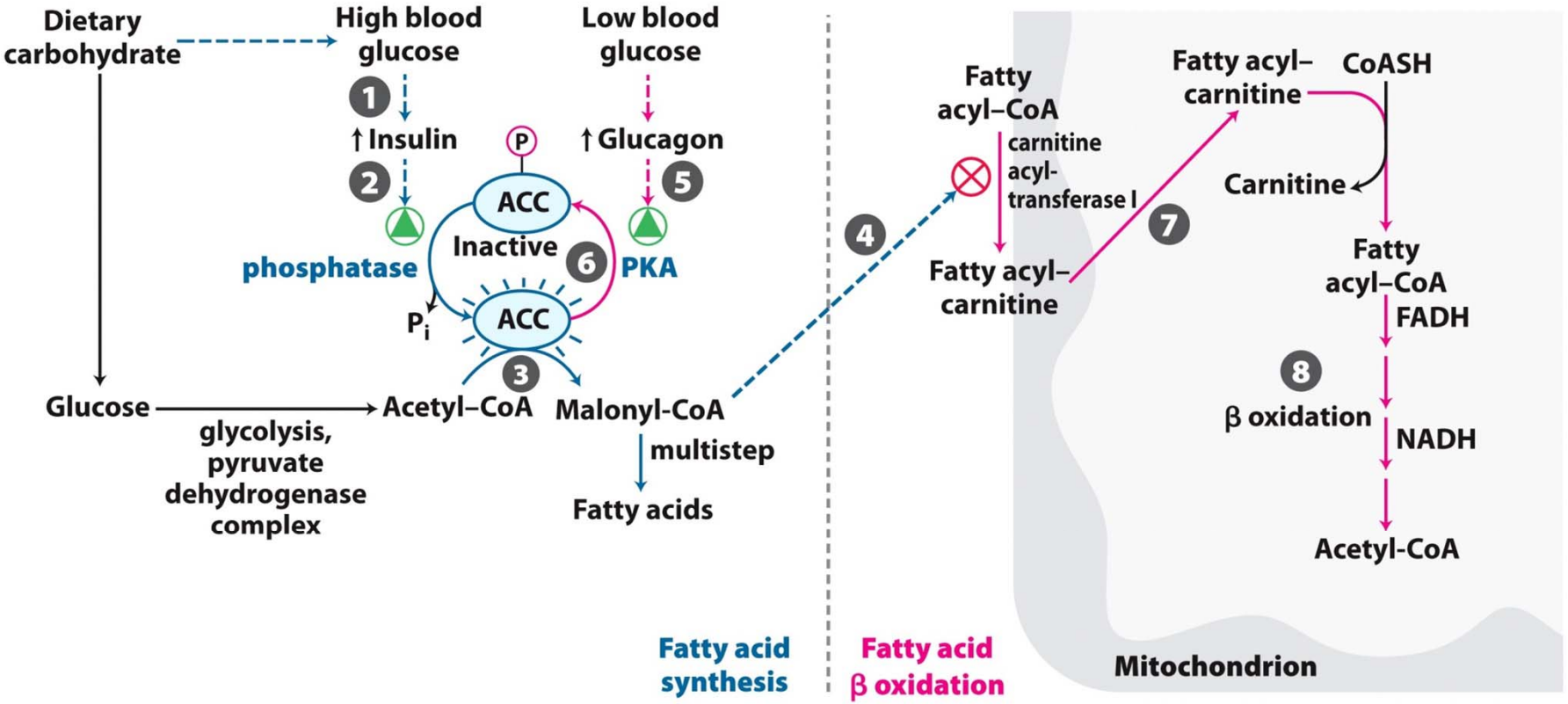


Figure 17-13
Lehninger Principles of Biochemistry, Sixth Edition
© 2013 W. H. Freeman and Company

Synthesis of Fatty Acids Is Catalyzed by Fatty Acid Synthase (FAS)

- Catalyzes a repeating four-step sequence that elongates the fatty acyl chain by two carbons at each step
- See Fig. 21-2
  - uses NADPH as as the electron donor
  - ises two enzyme-bound -SH groups as activating groups
- FAS I in vertebrates and fungi
- FAS II in plants and bacteria

The structure of fatty acid synthase type I systems

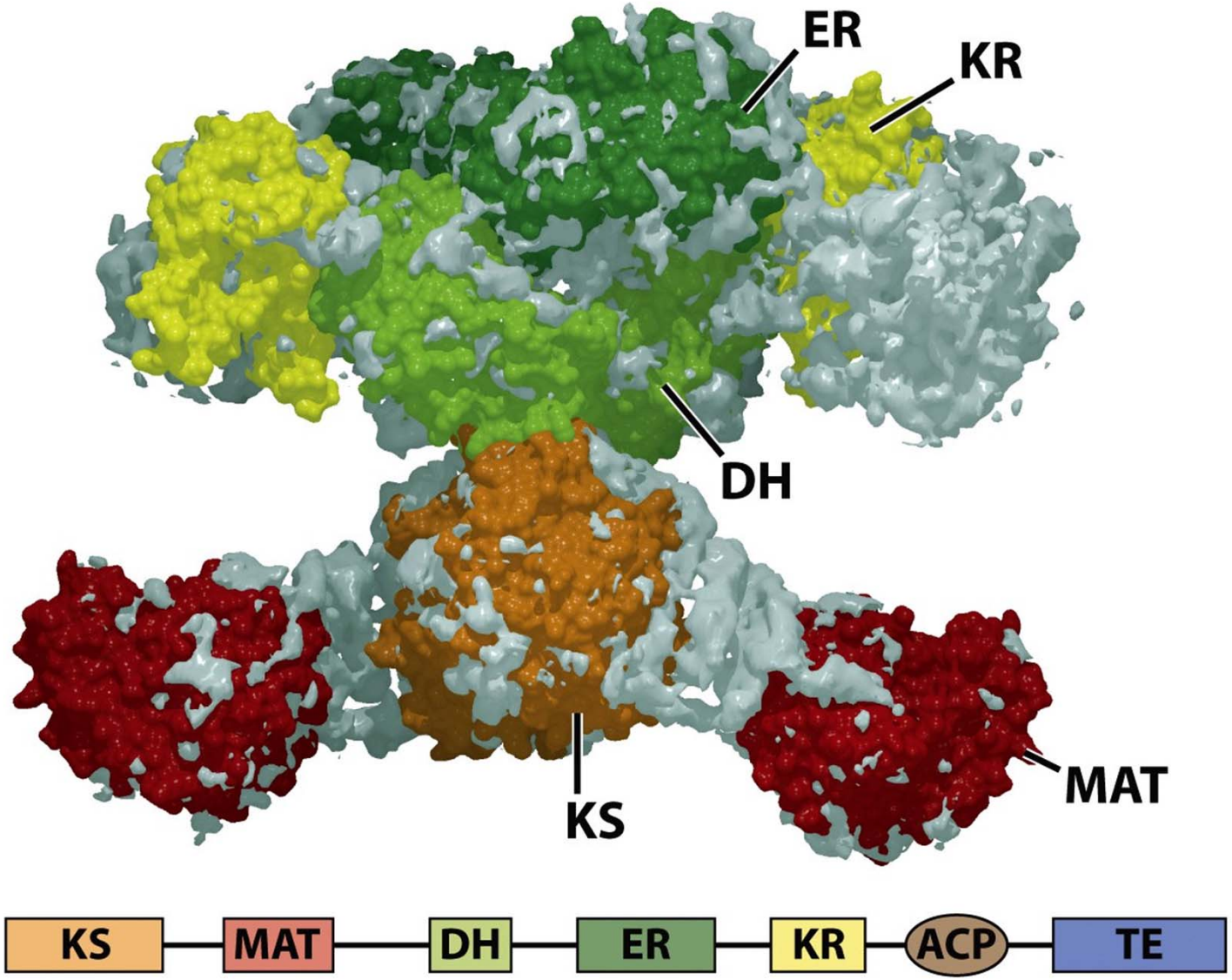


Figure 21-3a
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W. H. Freeman and Company

Acyl Carrier Protein (ACP) Serves as a Shuttle in Fatty Acid Synthesis

- Contains a covalently attached prosthetic group 4’-phosphopantetheine
  - flexible arm to tether acyl chain while carrying intermediates from one enzyme subunit to the next
- Delivers acetate (in the first step) or malonate (in all the next steps) to the fatty acid synthase
- Shuttles the growing chain from one active site to another during the four-step reaction

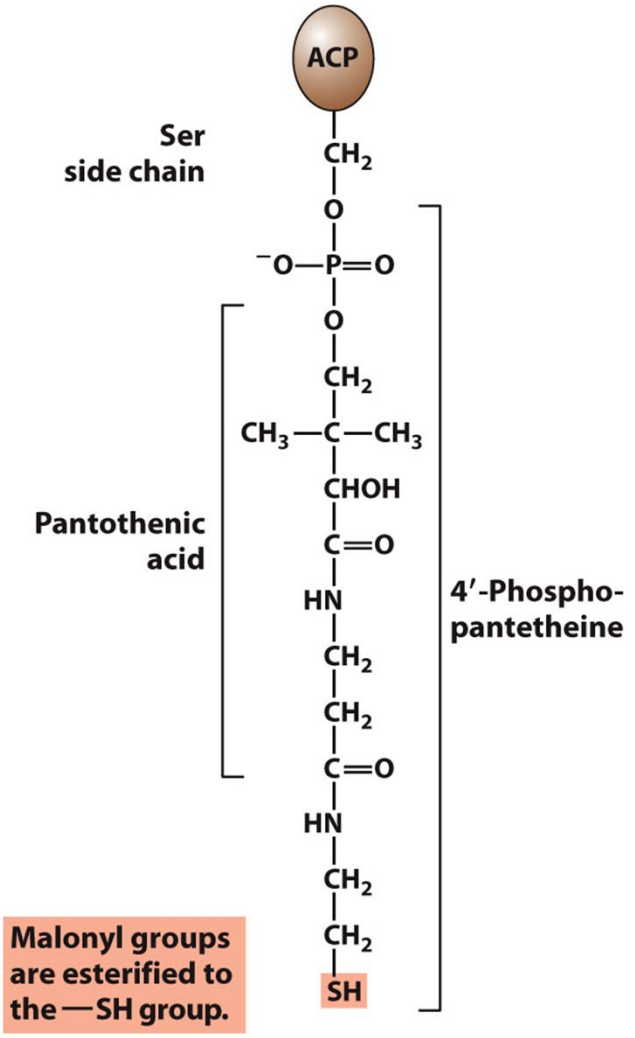


Figure 21-5
Lehninger Principles of Biochemistry, Seventh Edition
© 2017 W. H. Freeman and Company

Fatty Acid Synthesis

- Overall goal: attach acetate unit (2-carbon) from malonyl-CoA to a growing chain and then reduce it
- Reaction involves cycles of four enzyme-catalyzed steps:
  - condensation of the growing chain with activated acetate
  - reduction of carbonyl to hydroxyl
  - dehydration of alcohol to trans-alkene
  - reduction of alkene to alkane
- The growing chain is initially attached to the enzyme via a thioester linkage.

Sequence of events during synthesis of a fatty acid

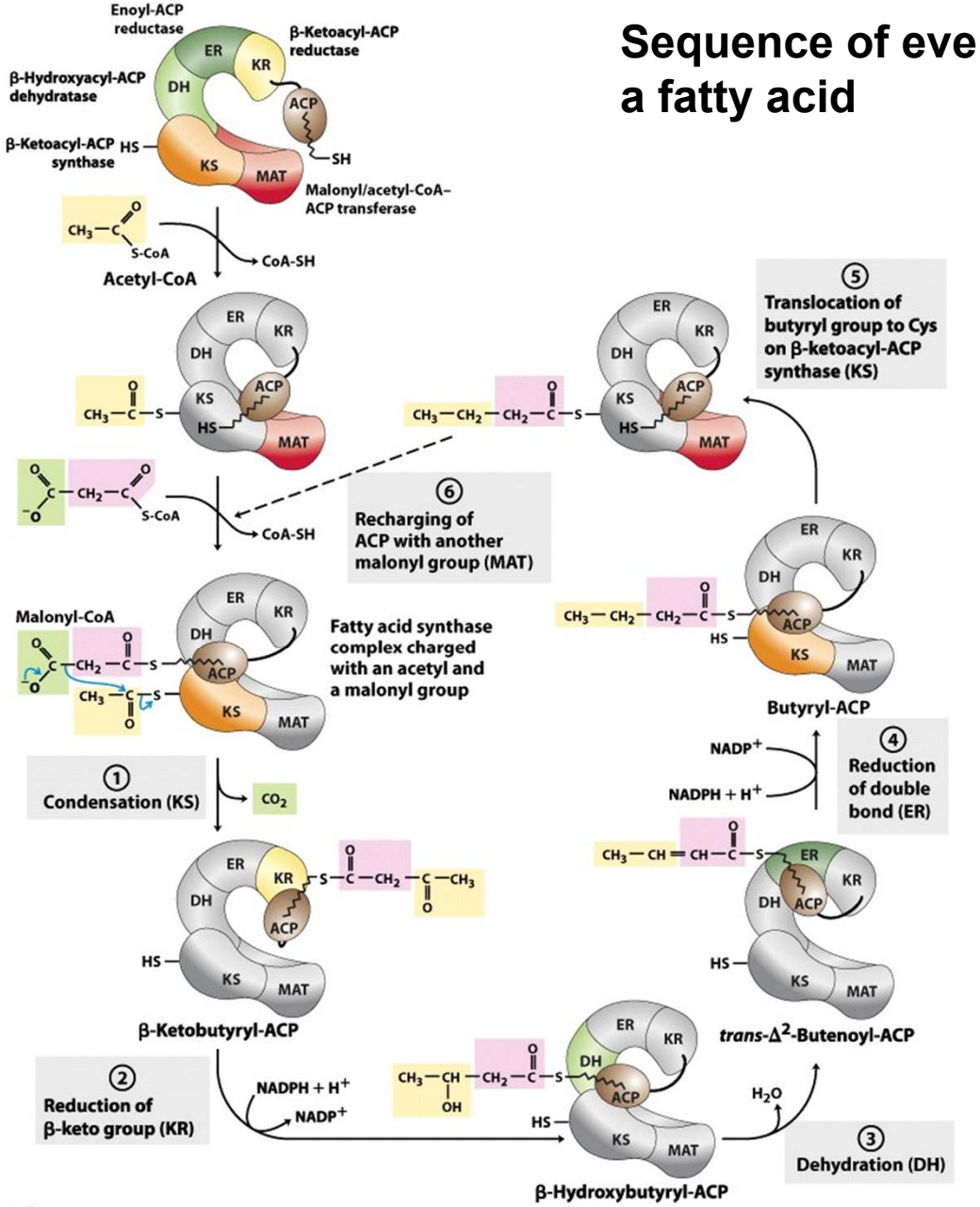


Figure 21-6

Lehninger Principles of Biochemistry, Fifth Edition

© 2008 W. H. Freeman and Company

Stoichiometry of Synthesis of Palmitate (16:0)

1. 7 acetyl-CoAs are carboxylated to make 7 malonyl-CoAs… using ATP.

7 acetyl-CoA + 7 CO_2 + 7 ATP → 7 malonyl-CoA + 7 ADP + 7 P_i

2. Seven cycles of condensation, reduction, dehydration, and reduction… using NADPH to reduce the β-keto group and trans-double bond

acetyl-CoA + 7 malonyl-CoA + 14 NADPH + 14 H^+ → palmitate (16C) + 7 CO_2 + 8 CoA + 14 $NADP^+$ + 6 H_2O

3. Addition of 1 and 2 leads

8 acetyl-CoA + 7 ATP +14 NADPH + 14 H^+ → palmitate + 8 CoA + 14 $NADP^+$ + 7 ADP+7Pi+6 H_2O

Palmitate Can Be Lengthened to Longer-Chain Fatty Acids

- Elongation systems in the endoplasmic reticulum and mitochondria create longer fatty acids.
- As in palmitate synthesis, each step adds units of 2 C.
- Stearate (18:0) is the most common product.

Palmitate and Stearate Can Be Desaturated

- Palmitate(16:0) → palmitoleate(16:1; Δ^9)
- Stearate (18:0) → oleate (18;1; Δ^9)
  - catalyzed by fatty acyl-CoA desaturase in animals
    - also known as the fatty acid desaturases
    - requires NADPH; enzyme uses cytochrome b_5 and cytochrome b_5 reductase

Note that this is a Δ^9-desaturase!

It reduces the bond between C-9 and C-10.

Plants Can Desaturate Positions Beyond C-9

- Humans have Δ^4, Δ^5, Δ^6, and Δ^9 desaturases but *cannot* desaturate beyond $\Delta^{9.}$
- Plants *can* produce:
  - linoleate 18:2($\Delta^{9,12}$)
  - α-linolenate 18:3 ($\Delta^{9,12,15}$)
- These fatty acids are “essential” to humans.
  - Polyunsaturated fatty acids (PUFAs) help control membrane fluidity.
  - PUFAs are precursors to eicosanoids.

Routes of synthesis of other fatty acids

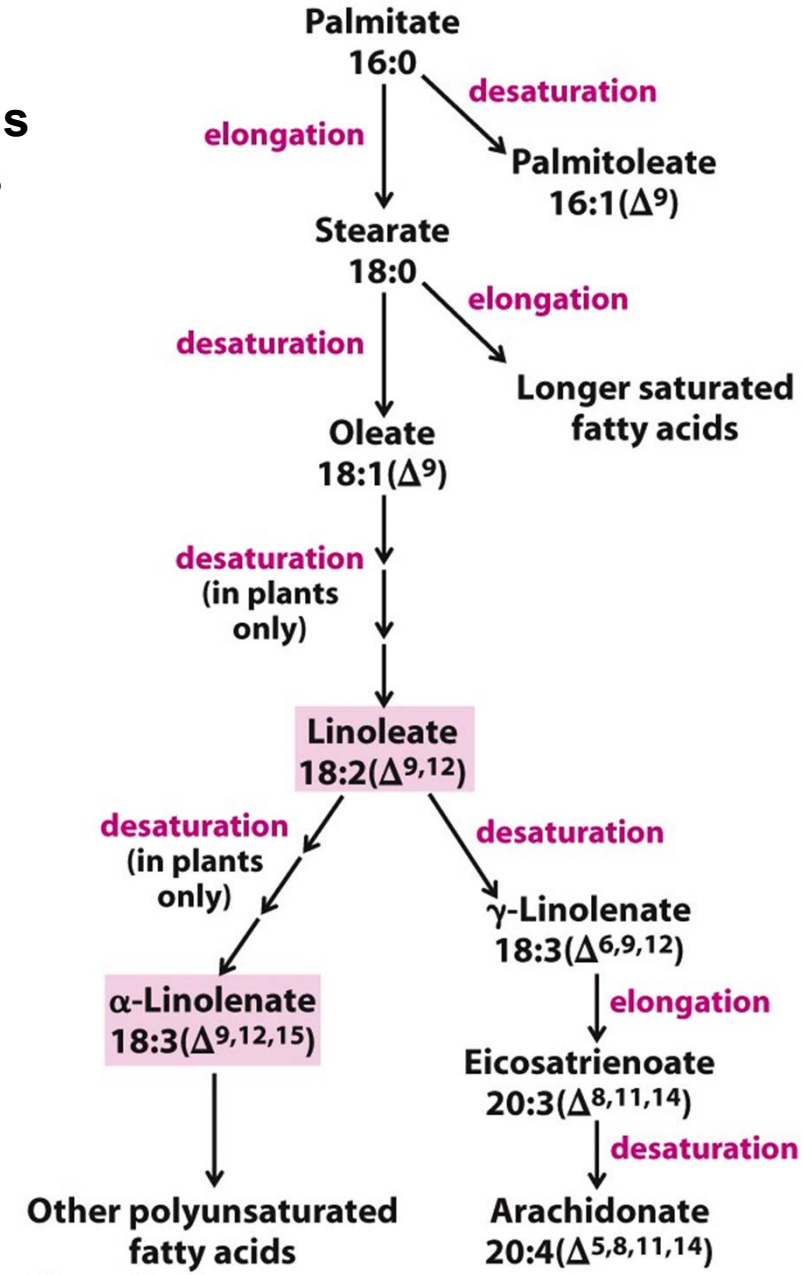


Figure 21-12

Lehninger Principles of Biochemistry, Fifth Edition

© 2008 W. H. Freeman and Company

Biosynthesis of TAG

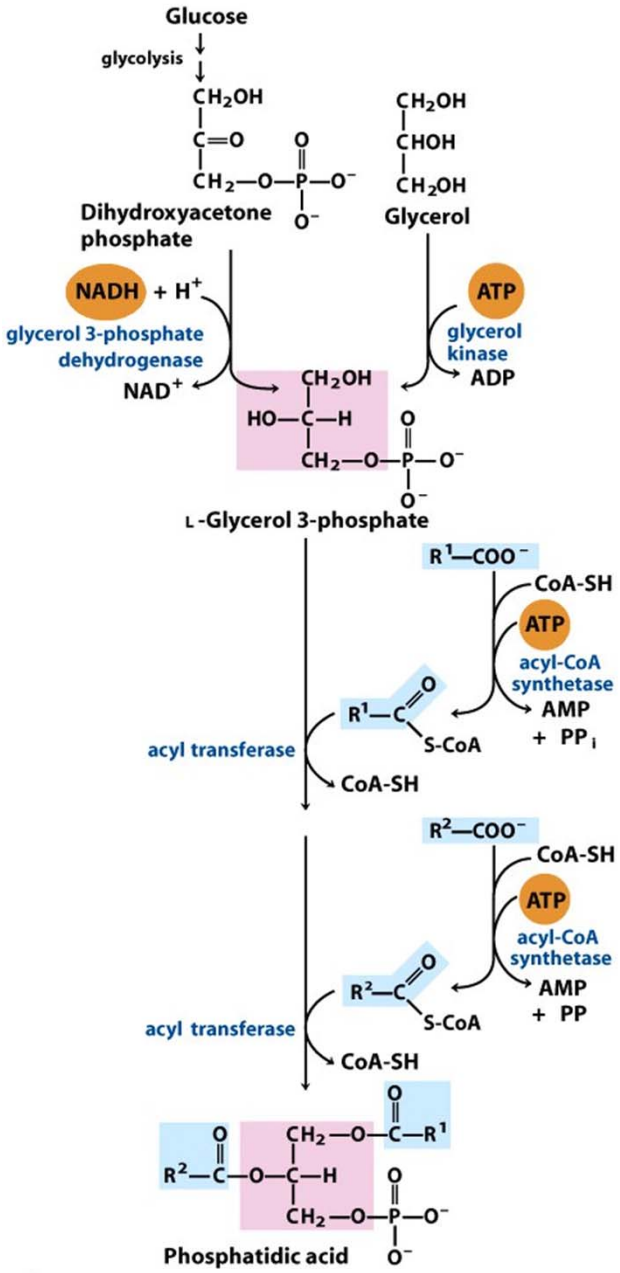


Figure 21-17

Lehninger Principles of Biochemistry, Fifth Edition

© 2008 W. H. Freeman and Company

Biosynthesis of TAG

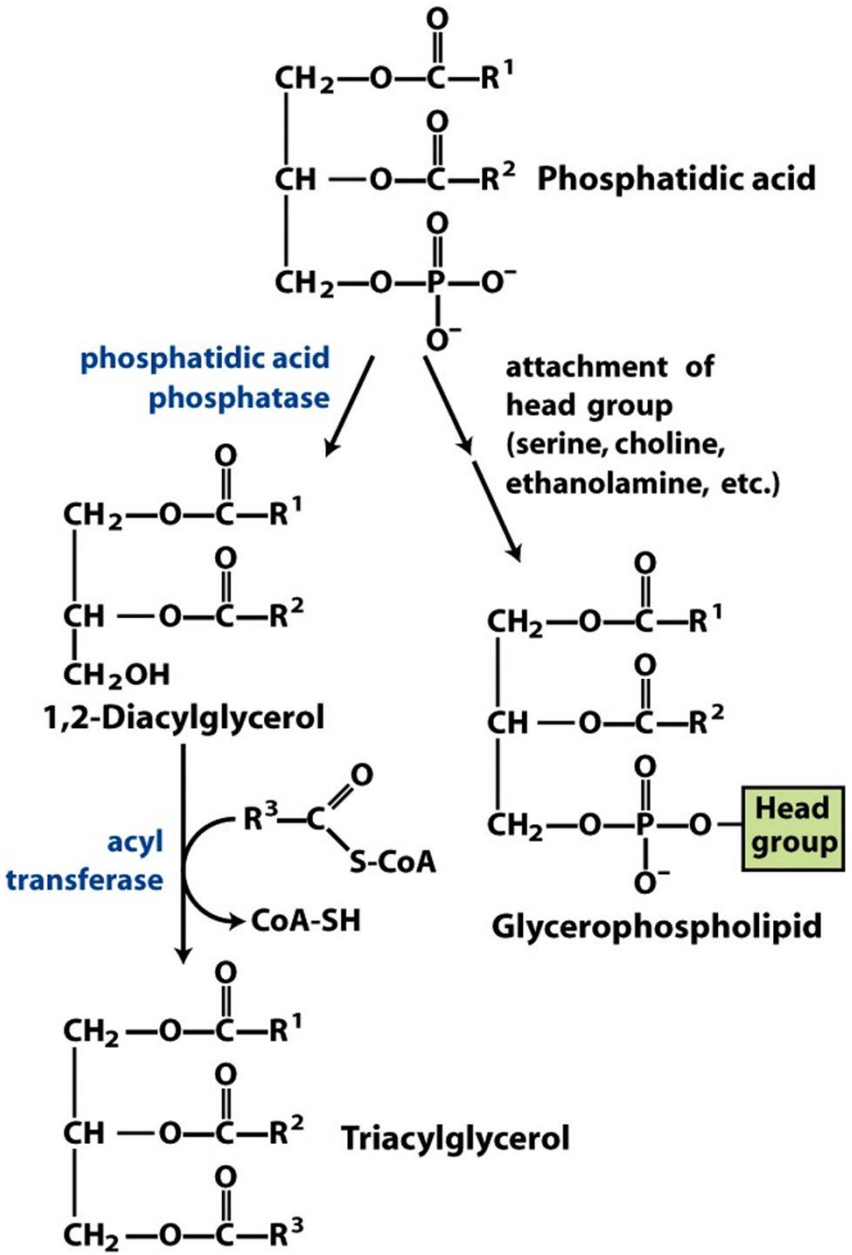


Figure 21-18
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W. H. Freeman and Company

Regulation of Triacylglycerol Synthesis by Insulin

- Insulin results in stimulation of triacylglycerol synthesis.
- Lack of insulin results in:
  - increased lipolysis
  - increased fatty acid oxidation
    - sometimes to ketones if citric acid cycle intermediates (oxaloacetate) that react with acetyl CoA are depleted
  - failure to synthesize fatty acids

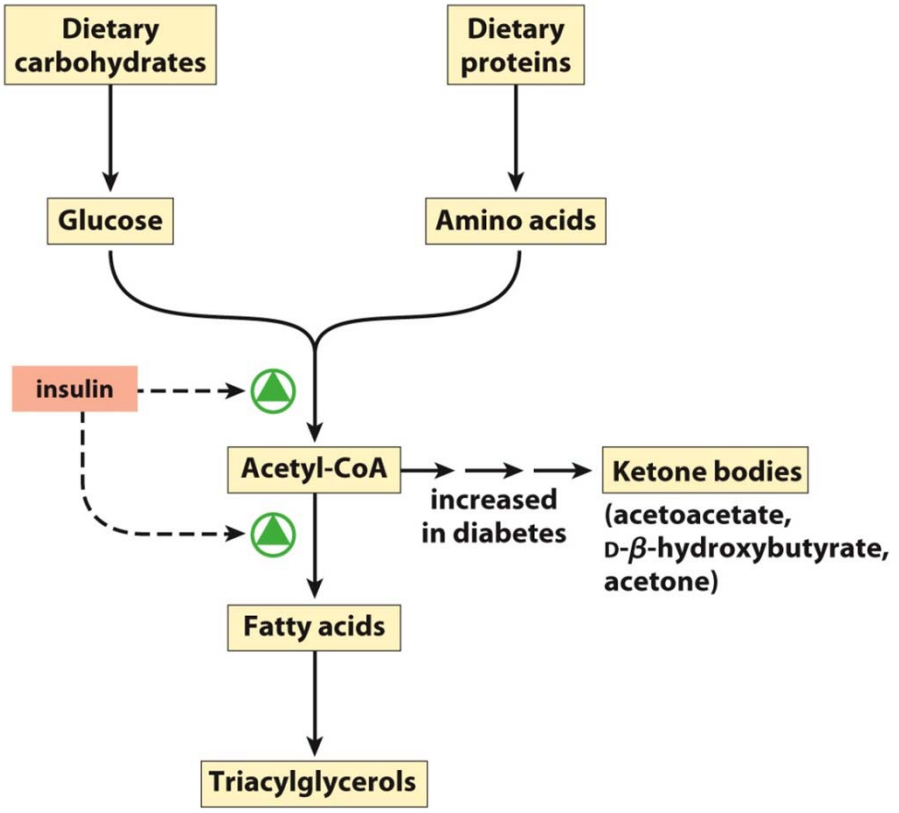


Figure 21-19
Lehninger Principles of Biochemistry, Seventh Edition
© 2017 W. H. Freeman and Company

Triacylglycerol Breakdown and Resynthesis create a Futile Cycle

- Seventy-five percent of free fatty acids (FFAs) released by lipolysis are reesterified to form TAGs, rather than be used for fuel.
  - Some recycling occurs in adipose tissue.
  - Some FFAs from adipose cells are transported to the liver, remade into TAG, and redeposited in adipose cells.
- Although the distribution between these two paths may vary, overall, the percentage of FFAs being esterified remains at ~75%.

The triacylglycerol cycle

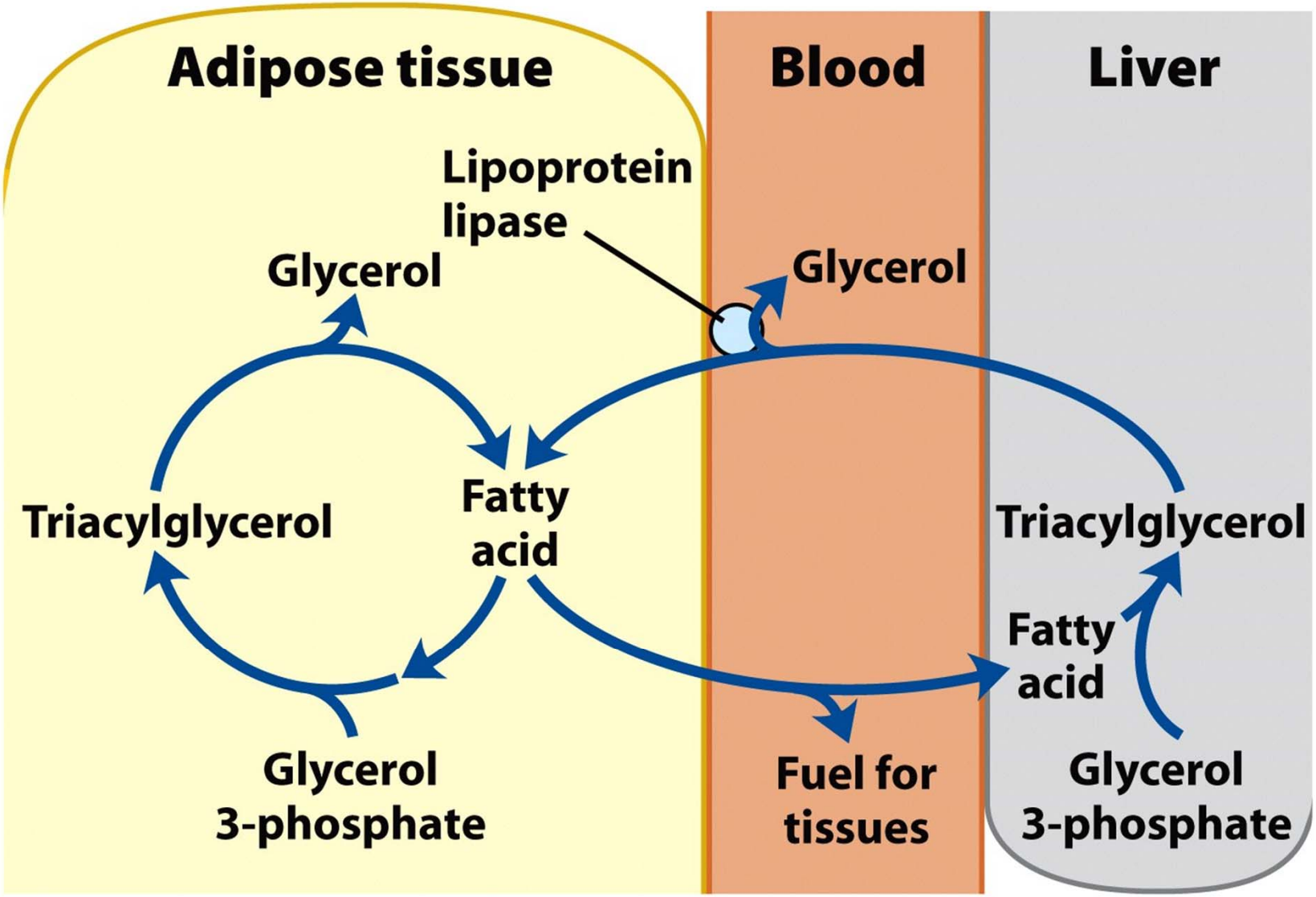


Figure 21-20
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W. H. Freeman and Company